

Ques. HOW CAN I GET STARTED AND STAY CONNECTED WITH ESP8266 WI-FI MODULE USING AT COMMANDS?

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Ans. ESP8266 Wi-Fi module is a self-contained system on chip with an integrated TCP/IP protocol stack that can give any microcontroller (MCU) access to your Wi-Fi network. It has now become a leading platform for the Internet of Things. This is a serial module with a built-in TCP/IP stack, so you can use it standalone.

You can make use of AT commands to connect with Wi-Fi networks and open TCP connections without having TCP/IP stack running in the MCU. Therefore you can connect any MCU to ESP module and start sending data to the Internet. In default mode, it boots up into serial module mode. In this mode you can start to communicate with it through a serial terminal using a set of AT commands that are supported by it.

Hardware connections. The hardware connections required to connect to ESP8266 module are fairly straightforward, but there are a couple of important items to note related to power. These are:

1. ESP8266 requires 3.3 volts power. Do not power it with 5 volts.
2. ESP8266 needs to communicate via serial at 3.3 volts and does not have 5-volt tolerant inputs, so you need level conversion to communicate with a 5-volt MCU like most Arduinos use.

With hardware connections in place you can communicate with the Wi-Fi module through a serial terminal. If you have Arduino IDE installed, click on the inbuilt serial monitor and follow the steps mentioned below:

1. Plug in the Wi-Fi module.
2. Choose the correct serial port through Tools > Serial Port menu.

TABLE I
SOME AT COMMANDS OF STANDARD FIRMWARE

ESP8266 Test	
AT+RST	Reset
AT+GMR	Check firmware version
AT+CWMODE=3	Set mode to AP+STA
AT+CWLAP	List Wi-Fi networks
AT+CWJAP="ssid", "password"	Connect to your Wi-Fi
AT+IOBAUD?	Check baud rate
AT+IOBAUD=9600	Set new baud rate

Q2. WHAT ARE THE DIFFERENCES BETWEEN DIFFERENT ARDUINO BOARDS? WHAT ARE THE DIFFERENCES BETWEEN ARDUINO UNO AND ARDUINO MEGA 2560?

Parmarthi Kanakraja

A2. All Arduino boards have one thing in common: these are programmed through Arduino IDE. Beyond that, there can be lot of differences. Number of inputs and outputs (the number of sensors, LEDs and buttons you can use

TABLE II
DIFFERENCES BETWEEN ARDUINO UNO AND ARDUINO MEGA 2560

Name	Processor	Analogue in/out	Digital I/O/PWM	EEPROM (kB)	SRAM (kB)	Flash (kB)	UART
Uno	ATmega328P	6/0	14/16	1	2	32	1
Mega 2560	ATmega2560	16/0	54/15	4	8	256	4

3. Ensure that Carriage Return is selected in the line ending pop-up menu at the bottom of the serial monitor. (For later versions it must be set to Both NL & CR.)

4. Ensure the communication speed is set to 115,200 baud. (For later versions it should be 9600 baud.)

You are now ready to send AT commands.

Some AT commands of standard firmware are enlisted in Table I.

For a detailed list of AT commands, you may refer to the following links:

<http://room-15.github.io/blog/2015/03/26/esp8266-at-command-reference/>
http://rancidbacon.com/files/kiwi-con8/ESP8266_WiFi_Module_Quick_Start_Guide_v_1.0.4.pdf

You may refer to the complete ESP8266 based project published in Electronics For You using the link given below:

<http://electronicsforu.com/electronics-projects/arduino-based-esp8266-wireless-web-server>

on a single board), speed, operating voltage and form factor are just a few variables.

Some boards are designed to be embedded and have no programming interface (hardware), which you would need to buy separately. Whereas, some can run directly from a 3.7-volt battery, others need at least 5 volts.

Refer Table II to find the right Arduino board for your project. Difference between Arduino Uno and Arduino Mega 2560 are mentioned there.

You may refer to the links given below to understand the differences between Arduino Uno and Arduino Mega 2560.

<https://www.sparkfun.com/products/11021>

<https://nettigo.eu/products/arduino-mega-2560-rev3>

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